Cover letter

Journal of Number Theory 25 October 2026

Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “The optimal log log q bound in the Pólya-Vinogradov inequality” by Deep Bhattacharjee and Ushashi Bhattacharya.

The unchanged article derives the optimal-order character-sum bound from the credited zero-free theorem. The research supplement supplies a separate full formal-source ZIP and studies a sufficient mean-square input for a weaker 15/16 boundary. That input is not proved; no independent zero-free replacement is claimed.

The character-sum, Dirichlet L-function and logarithmic-derivative results fit analytic number theory and the journal’s detailed technical-article scope.

The prior public preprint is `https://hexagonmath.org/2610.00256v2`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23629649`. Editable sources and verification records are supplied with the manuscript.

The unchanged proof tree has the completed 19 original, seven endpoint and 20 component axiom reports, using only propext, Classical.choice and Quot.sound. Source identity and historical receipts were rechecked. No new full Lean build or independent zero-free proof is claimed; the new conditional research note is not formalized.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The earlier archived manuscript had a human-understanding confirmation. Added research material awaits author review. The named human authors retain responsibility for the content.

Sincerely,

Deep Bhattacharjee
Corresponding author
itsdeep@live.com
ORCID: 0000-0003-0466-750X